GLUCOTWIN AI

System Architecture & Technical Overview

A digital-twin-inspired educational prototype exploring patient-state representation and short-horizon glucose forecasting using synthetic EHR data and CGM time series.

PROJECT SUMMARY

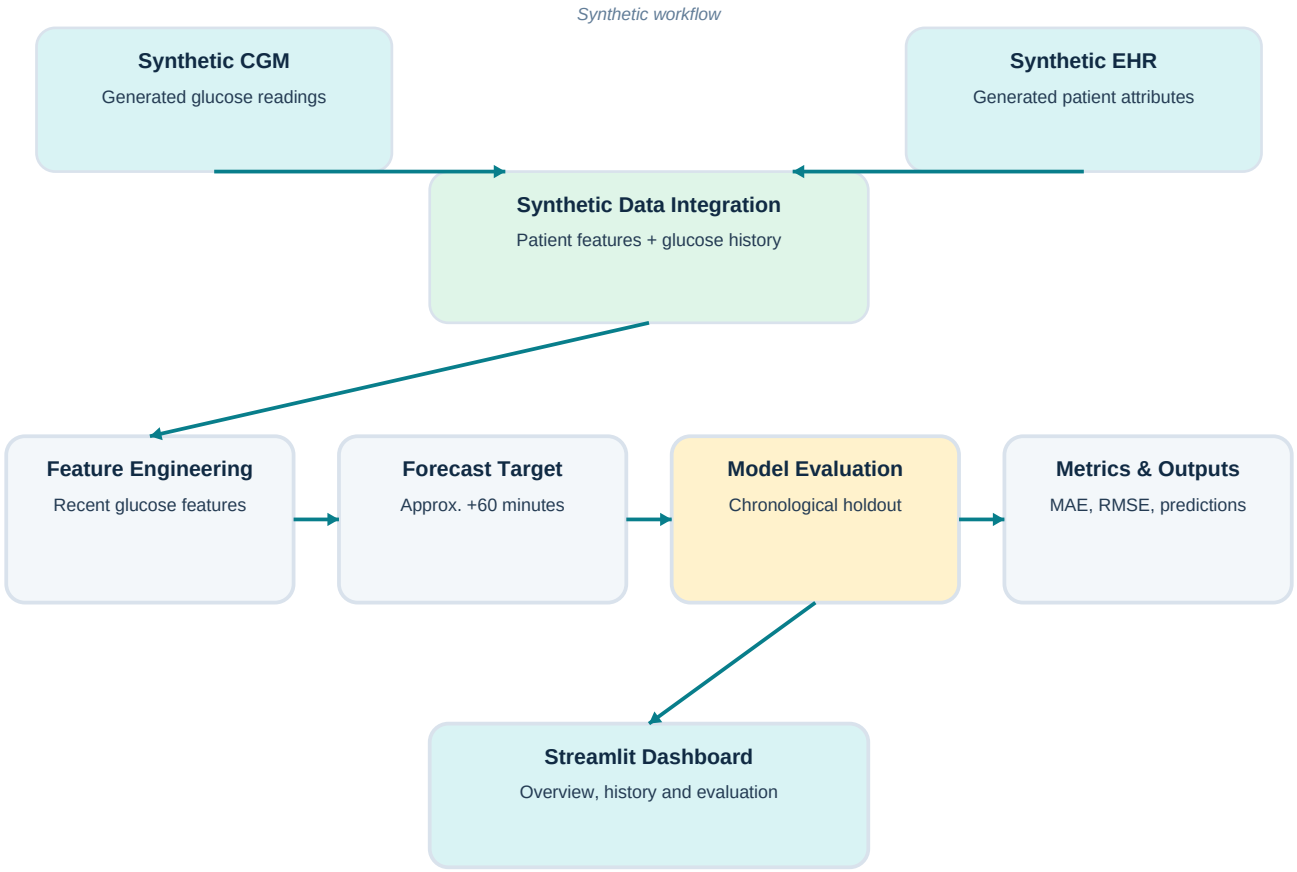
GlucoTwin AI combines data preparation, time-series feature engineering, regression-based forecasting, model evaluation, and an interactive Streamlit dashboard. It presents a synthetic patient workflow separately from a real CGM experiment involving one participant.

Field	Details
Project	GlucoTwin AI
Author	Molly Khandelwal
Programming language	Python
Dashboard	Streamlit
Forecast horizon	Approximately 60 minutes
Repository	github.com/MollyKhandelwal/GlucoTwin-AI
Document generated	2026-10-18

Important: This is a research and educational prototype. It is not a medical device and must not be used for diagnosis, treatment, or clinical decision-making.

1. System Architecture

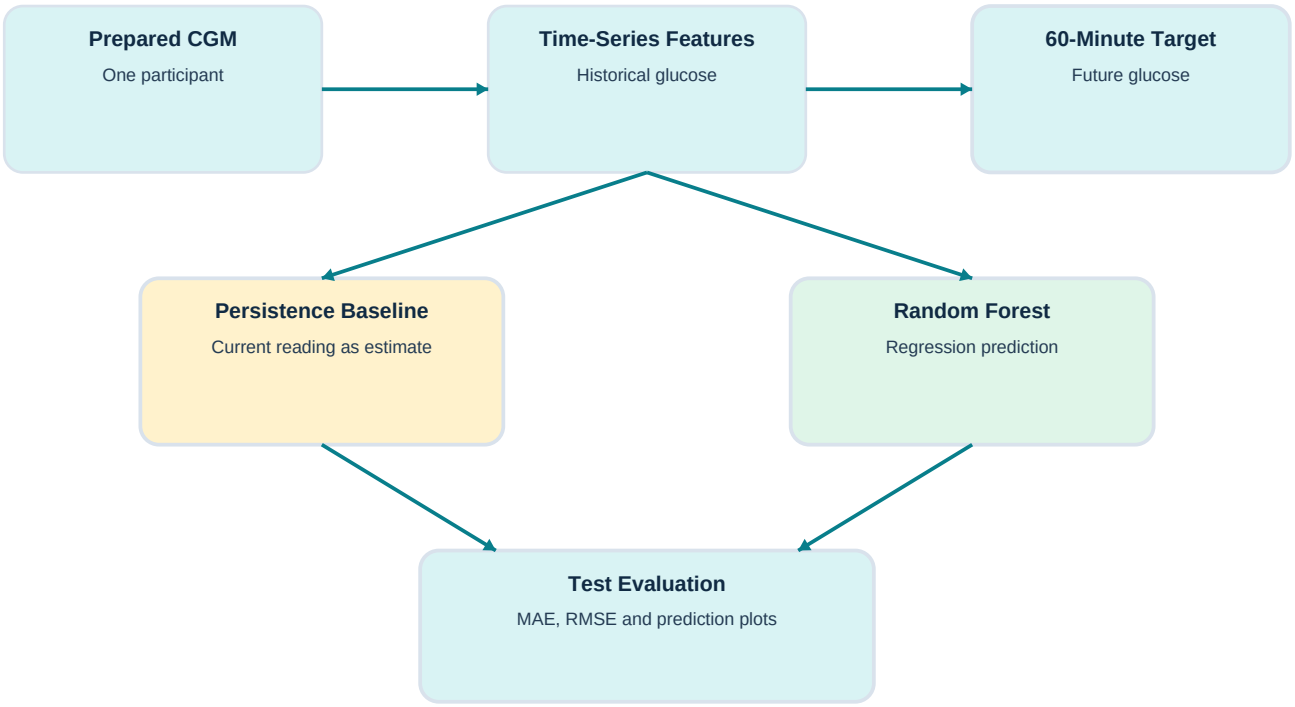
The synthetic workflow demonstrates the integration of generated patient attributes and generated glucose readings. The resulting features feed a forecasting and evaluation pipeline, with saved outputs displayed in the dashboard.



The diagram is a logical view of the implemented prototype. The synthetic workflow and real CGM workflow remain separate; the diagram does not imply that synthetic EHR records belong to the real CGM participant.

2. Real CGM Experiment

The real-data workflow uses prepared Dexcom CGM readings from one participant. Historical glucose values are used to construct forecasting features and an approximately 60-minute target. A persistence baseline and a Random Forest regressor are evaluated on the same recorded test samples.



Recorded test metrics

Model	MAE (mg/dL)	RMSE (mg/dL)
Persistence baseline	9.431	14.156
Random Forest	10.266	14.152

Interpretation: Random Forest has higher MAE and marginally lower RMSE than the persistence baseline. It does not outperform the baseline on both metrics.

These results are exploratory and apply only to the evaluated participant and test period. They do not establish generalization to other participants or clinical accuracy.

3. Components and Responsibilities

Component	Responsibility
Data preparation	Loads and prepares synthetic data or prepared CGM readings.
Synthetic EHR	Provides generated patient attributes for the synthetic experiment.
Feature engineering	Creates historical glucose features and forecasting targets.
Persistence baseline	Uses the current glucose value as the future estimate.
Random Forest	Fits a regression model to historical glucose features.
Evaluation	Calculates MAE and RMSE on held-out test samples.
Outputs	Stores metrics, predictions, and visualizations.
Streamlit dashboard	Displays patient overview, time series, forecasts, and evaluation.

4. Technology Stack

Technology	Role
Python	Core programming language
Pandas and NumPy	Data processing and feature preparation
Scikit-learn	Random Forest regression and evaluation
Matplotlib	Forecast and error visualization
Streamlit	Interactive dashboard
CSV	Data and experiment output storage
Git and GitHub	Version control and project hosting

5. Forecasting and Evaluation

1. Load data: Read the relevant synthetic or prepared CGM dataset.

2. Prepare timestamps: Inspect ordering, missing values, and sampling intervals.

3. Build features: Construct historical glucose features.

4. Define the target: Estimate glucose approximately 60 minutes ahead.

5. Split chronologically: Keep later observations for testing.

6. Evaluate: Compare persistence and Random Forest predictions.

7. Save results: Store metrics, predictions, and plots.

8. Visualize: Present experiment outputs in Streamlit.

Evaluation metrics

Metric / Method	Explanation
MAE	Mean absolute error between predicted and observed glucose.
RMSE	Root mean squared error; larger errors contribute more heavily.
Persistence baseline	Predicts future glucose using the current reading.

6. Limitations and Responsible Use

• Synthetic EHR and CGM records are generated for demonstration.

• The real CGM experiment contains data from one participant.

• Synthetic EHR records are not a clinically matched record for the real CGM participant.

• Single-participant performance does not establish population-level generalization.

• External validation on additional participants is required.

• The dashboard must not be used to make medical decisions.

7. Future Improvements

• Evaluate on multiple independent participants.

• Improve time-series validation and model selection.

• Explore prediction intervals and uncertainty estimation.

• Improve missing-data handling and robustness.

• Conduct rigorous external validation before any clinical application.

GlucoTwin AI is an educational proof of concept. Its current results do not demonstrate clinical validity.

Project Repository

https://github.com/MollyKhandelwal/GlucoTwin-AI